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Q1. Deadlock Debugging

Problem Identification:

The deadlock occurs because P1 locks R1 then waits for R2, while P2 locks R2 then waits for R1. Each process holds one resource while waiting for the other.

Debugging (Root Cause):

1. P1 acquires R1.
2. P2 acquires R2.
3. P1 requests R2 (blocked).
4. P2 requests R1 (blocked).

This creates a circular wait causing deadlock.

Deadlock Conditions:

- Mutual Exclusion
- Hold and Wait
- No Preemption
- Circular Wait

Optimization:

Use a fixed resource ordering:

P1: lock(R1) → lock(R2)

P2: lock(R1) → lock(R2)

Expected Result:

Deadlock is eliminated because circular wait is removed.

Q2. CPU Utilisation Problem

Problem Identification:

CPU utilization is only 30% even though processes are ready.

Debugging (Root Cause):

The scheduler repeatedly shows Ready → Waiting → Ready. This indicates I/O-bound processes that spend more time waiting for I/O than using the CPU.

Optimization:

1. Increase the degree of multiprogramming.
2. Use asynchronous or faster I/O devices.

Expected Result:

CPU remains busy while other processes wait for I/O, improving utilization and throughput.

Q3. Memory Bottleneck Debugging

Problem Identification:

Memory usage:

- Code = 5 MB
- Heap = 150 MB
- Stack = $25 \times 1 \text{ MB} = 25 \text{ MB}$

Total = 180 MB.

Debugging (Root Cause):

Although memory fits within 2 GB RAM, frequent page faults occur because many threads access different memory pages, reducing memory locality.

Optimization:

1. Optimize heap memory by freeing unused objects and using memory pools.
2. Use a thread pool or reduce stack size to lower memory pressure.

Expected Result:

Fewer page faults, improved memory utilization, and faster execution.

Q4. Starvation Debugging

Problem Identification:

Process P5 in Q3 never receives CPU time.

Debugging (Root Cause):

The scheduler always gives priority to Q1 and Q2. Continuous arrival of higher-priority processes causes indefinite postponement of P5 (starvation).

Optimization:

Implement Aging by gradually increasing the priority of long-waiting processes.

Expected Result:

Every process eventually receives CPU time while maintaining fairness.

Q5. Context-Switch Overhead Debugging

Problem Identification:

The server performs 20,000 context switches per second.

Debugging (Root Cause):

Average burst time is 2 ms but the time quantum is only 1 ms, causing frequent preemption. High I/O wait also increases context switches.

Optimization:

1. Increase the time quantum to 4–5 ms.
2. Reduce I/O blocking using asynchronous or buffered I/O.

Expected Result:

Context switches decrease, CPU overhead is reduced, and system performance improves.